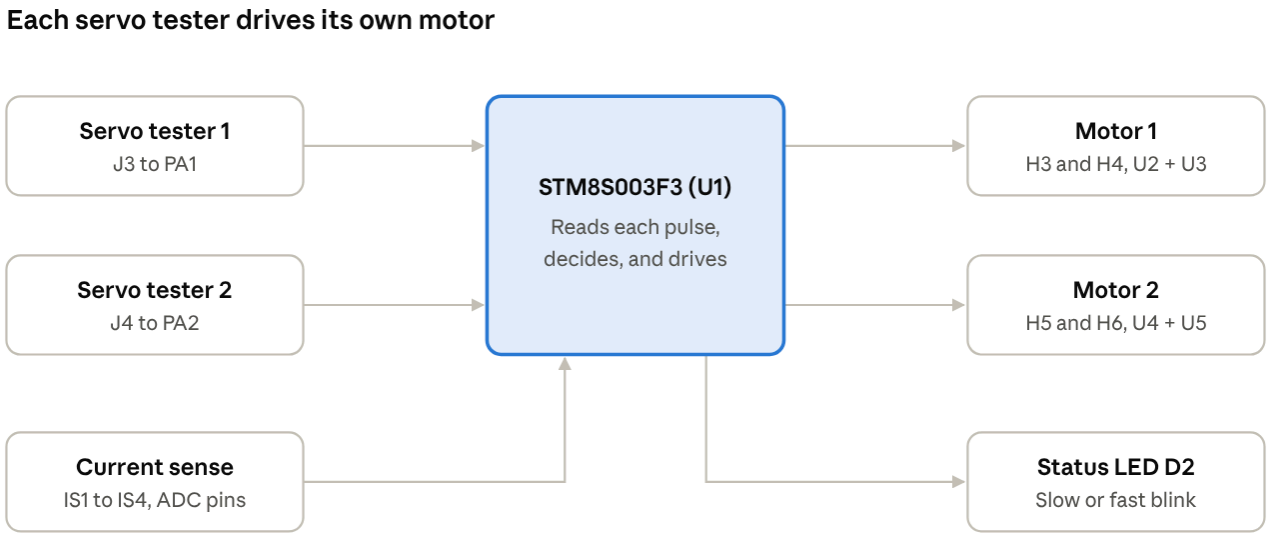
20A DC Controller: Servo Input to Motor Control

Firmware ServoMotor_Fault.ino for the STM8S003F3U6 board (sduino, SDCC). Written against schematic 20A_DC_Controller_V1. Not yet verified on hardware.

1. Overview

Two servo testers feed the board. The tester on J3 controls Motor 1 and the tester on J4 controls Motor 2. Each tester sends one pulse about every 20 ms, and the width of that pulse (about 1 to 2 ms) sets the motor direction and speed. The status LED shows whether the board is healthy (slow blink) or has detected an overcurrent fault (fast blink). There is no serial output, because the board has no free serial port.



system overview · 2 inputs, 2 motors

Each tester feeds its own pin and motor. Current sense feeds back into the STM8, and the status LED reports the result.

2. Pin map

Function	Pin no.	MCU pin	Notes
Servo input 1	2	PA1	J3 pin 1, through R5 (1K)
Servo input 2	3	PA2	J4 pin 1, through R9 (1K)

Function	Pin no.	MCU pin	Notes
Motor 1, side A (U2)	13	PC6	High for forward, low for reverse
Motor 1, side B (U3)	18	PD4	Carries the PWM
Motor 2, side A (U4)	14	PC7	High for forward, low for reverse
Motor 2, side B (U5)	7	PA3	Carries the PWM
INH enable	10	PC3	Shared enable for U2 to U5
Status LED D2	12	PC5	Active high, through R23 (1K)
Current sense IS1 (U2)	11	PC4	ADC input
Current sense IS2 (U3)	17	PD3	ADC input
Current sense IS3 (U4)	16	PD2	ADC input
Current sense IS4 (U5)	20	PD6	ADC input, enabled because serial is not used
Voltage_Sense	19	PD5	Check disabled by default (see section 9)

Motor pads: Motor 1 is on H3 and H4, Motor 2 is on H5 and H6. The motor supply goes on H1 (VCC) and H2 (PGND).

3. The servo tester signal

The servo tester (HW-141 board) contains a potentiometer, the knob. The knob only changes a resistance. A small circuit on the board turns that setting into a servo-style pulse: a pulse about every 20 ms (50 Hz) whose width is typically 1000 to 2000 microseconds. The STM8 never sees the knob itself. It only measures the width of the pulse.

The tester is powered from the board: J3 and J4 pin 2 carry +5V and pin 3 is GND. The pulse range here is the typical servo range and has not been measured on your tester.

4. Pulse width to motor command

Pulse width	Meaning	Motor
1000 to 1450 us	Knob low	Reverse, speed falls from 100% to 0% as the pulse rises
1450 to 1550 us	Dead zone	Stop (brake)
1550 to 2000 us	Knob high	Forward, speed rises from 0% to 100%
Below 800 us, above 2200 us, or no pulse	Invalid	That motor stops

The dead zone stops the motor from creeping because of small errors in the pulse. Each input is handled separately, so a missing signal on one input stops only its own motor.

5. How each motor is driven

Each BTN9970LV (U2 to U5) is one half-bridge. Two of them make an H-bridge for one motor. With INH high, an IN pin high connects that output to VCC, and an IN pin low connects it to ground.

PWM is placed on the B pins (PD4 and PA3), which are the default timer-2 PWM pins, so no option-byte remap is needed. The A pins are plain high or low outputs.

Command	Side A (PC6 / PC7)	Side B (PD4 / PA3)	Result
Forward	High	PWM, low for the on-time (duty 255 minus speed)	Motor turns forward at the set speed
Reverse	Low	PWM, high for the on-time (duty equals speed)	Motor turns in reverse
Stop	Low	Low	Motor brakes

INH (PC3) is high when at least one input has a valid pulse. It goes low only when both inputs have no valid pulse, which switches all four drivers off.

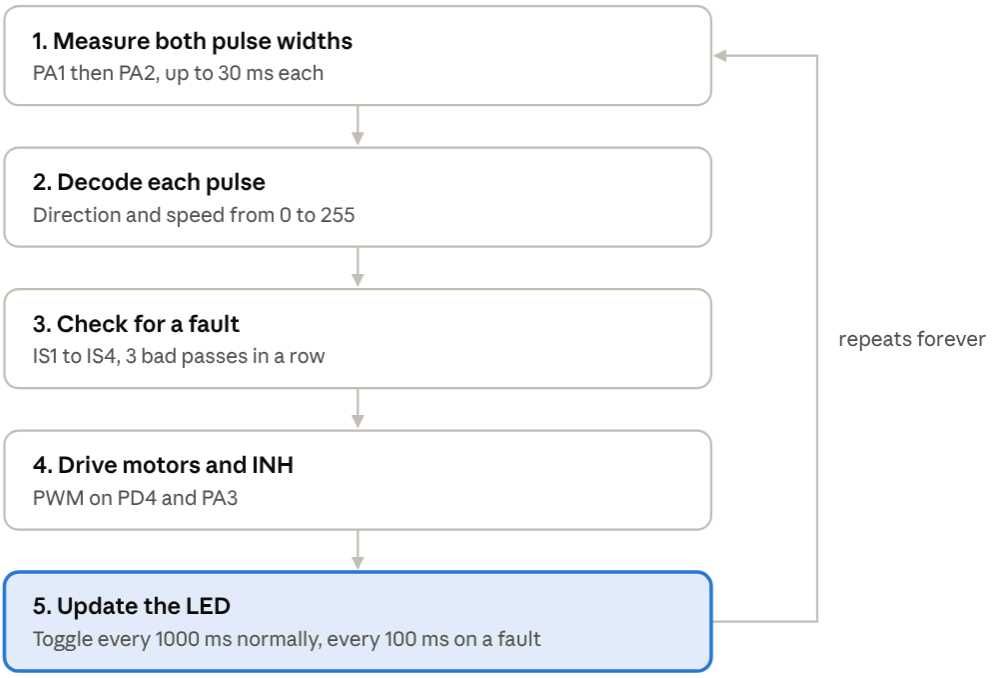
6. The main loop

loop() repeats these five steps forever:

1. Measure the pulse width on PA1, then on PA2. Each measurement waits up to 30 ms.
2. Decode each pulse into a direction and a speed from 0 to 255.
3. Check the current-sense pins for a fault.
4. Drive both motors and set INH.
5. Update the status LED.

Because the code waits for two pulses, one pass takes roughly 40 ms when both testers are connected. Each motor is therefore updated about 25 times per second, which is enough for a knob.

The loop runs five steps, then repeats



main loop · 5 steps, repeats

The LED update is the last step, so it runs once at the end of every pass.

7. Fault detection

checkFault() reads IS1 to IS4. If any reading is above 600 ADC counts (about 2.9 V at the pin with a 5V reference), that pass counts as bad. A fault is declared only after 3 bad passes in a row, which filters out brief noise from the PWM switching. One clean pass clears the fault.

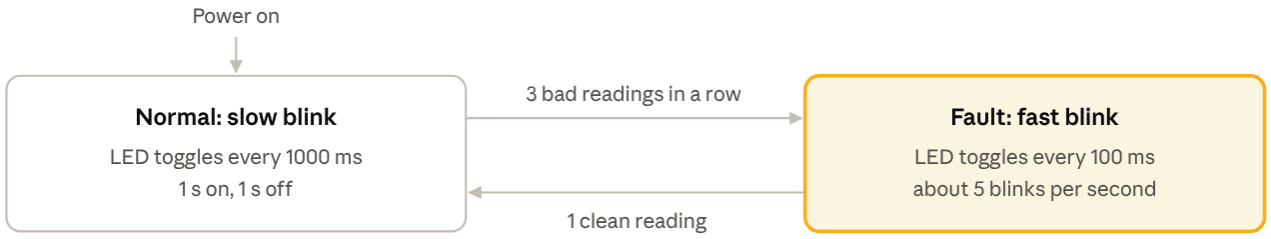
What a fault does depends on the FAULT_SHUTDOWN setting. It is 0 by default, so a fault only changes the LED and does not stop the motors. Set it to 1 after the 600-count threshold has been calibrated, and a fault will also switch the drivers off.

8. LED pattern

The status LED D2 on PC5 is the only status output on the board.

State	On time	Off time	Blink rate	Meaning
Normal	1000 ms	1000 ms	One blink every 2 seconds	Firmware is running and no fault is present
Fault	100 ms	100 ms	About 5 blinks per second	Overcurrent seen on IS1 to IS4 for 3 passes in a row
Off or steady	n/a	n/a	None	MCU is not running, is unpowered, or is not programmed

The LED blinks slowly until a fault speeds it up



LED states · 2 states, 2 transitions

The board starts in the slow state. Three bad current readings in a row switch the LED to the fast blink, and one clean reading switches it back.

How the pattern is produced:

1. updateLed() runs once at the end of every loop pass.
2. It picks an interval: 1000 ms normally, or 100 ms when a fault is active.
3. It compares the interval with the time since the last toggle, using millis().
4. If enough time has passed, it flips the LED and records the time.

The LED code never waits, so it does not delay the motor control. The check only happens between loop passes, so the real toggle times round up to the next pass. With a pass of about 40 ms, the fast blink is really about 100 to 120 ms.

The LED does not show a missing input signal. If a tester is unplugged, its motor stops and the LED keeps its normal slow blink.

9. Things to verify before relying on this

- Voltage divider: R16 (10K on top) and R18 (43K to ground) put about 81% of VCC on PD5. Swap the two values before applying VCC. Only then should the voltage check (ENABLE_VSENSE_CHECK) be switched on.
- Fault threshold: 600 counts is a placeholder. Measure the IS voltage at normal and maximum motor current and set IS_FAULT_COUNTS just above the normal level.
- Pulse range: 1000 to 2000 us is the typical servo range. Measure your tester and adjust the neutral point and dead zone if the motors creep.
- Motor 2 hardware: a weak or one-way motor 2 was traced to the U4 and U5 drivers. That is a hardware issue the firmware cannot fix.
- PWM pins: PD4 and PA3 are assumed to be supported by analogWrite in your sduino variant. If a motor does not respond to speed changes, test it first with digitalWrite.
- Flash: the STM8S003F3 has 8 KB. Removing serial output saved most of the space, but a ROM overflow error would mean the float code or other features need to be trimmed.